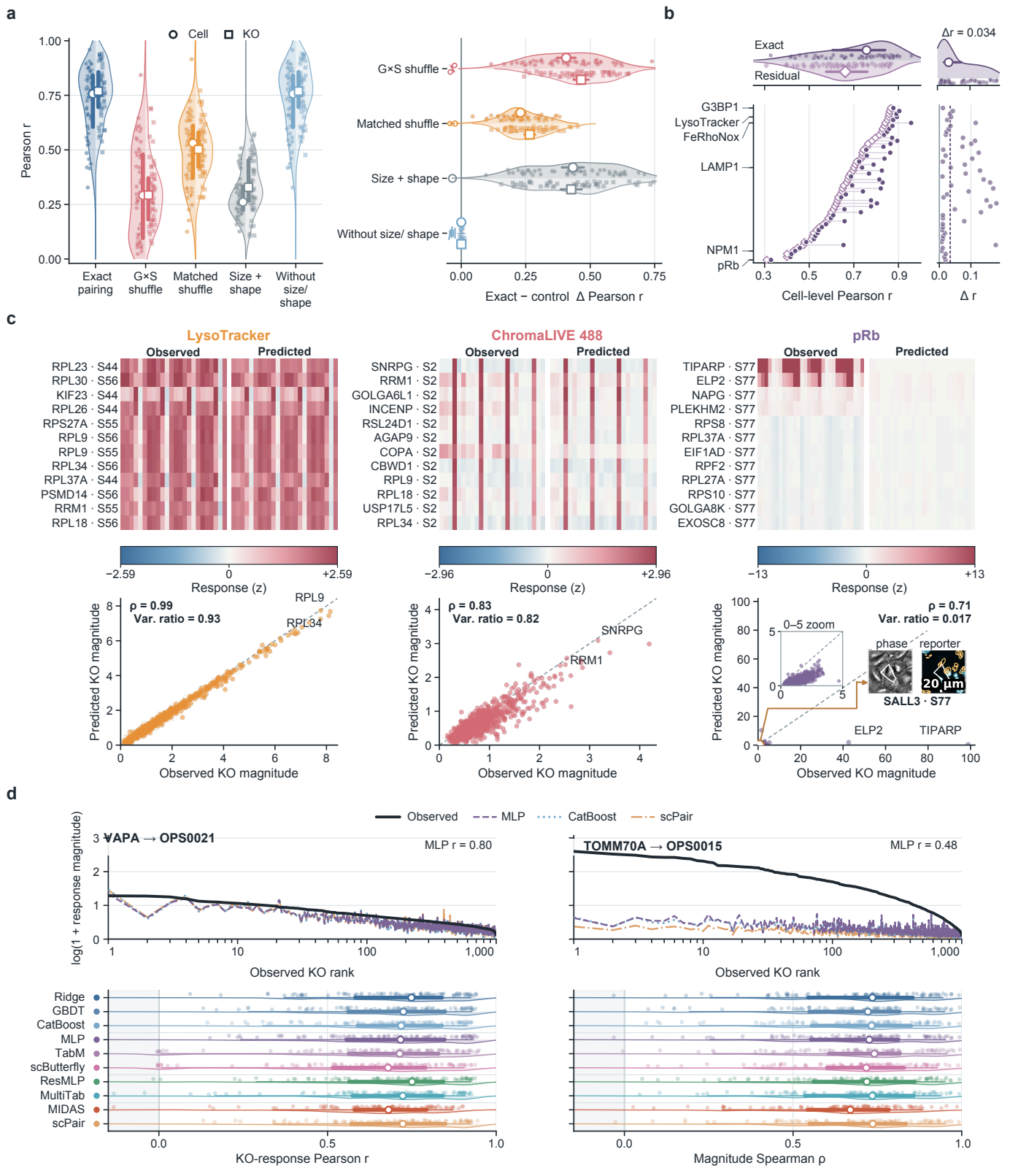

a
Cell
KO
Pearson r
1.00
0.75
0.50
0.25
0.00
Exact pairing
G×S shuffle
Matched shuffle
Size + shape
Without size/ shape
G×S shuffle
Matched shuffle
Size + shape
Without size/ shape
0.00
0.25
0.50
0.75
Exact − control Δ Pearson r
b
Exact
Residual
Δr = 0.034
G3BP1
LysoTracker
FeRhoNox
LAMP1
NPM1
pRb
0.3
0.5
0.7
0.9
Cell-level Pearson r
0.0
0.1
Δ r
c
LysoTracker
ChromaLIVE 488
pRb
Observed
Predicted
RPL23 · S44
RPL30 · S56
KIF23 · S44
RPL26 · S44
RPS27A · S55
RPL9 · S56
RPL9 · S55
RPL34 · S56
RPL37A · S44
PSMD14 · S56
RRM1 · S55
RPL18 · S56
SNRPG · S2
RRM1 · S2
GOLGA6L1 · S2
INCENP · S2
RSL24D1 · S2
AGAP9 · S2
COPA · S2
CBWD1 · S2
RPL9 · S2
RPL18 · S2
USP17L5 · S2
RPL34 · S2
TIPARP · S77
ELP2 · S77
NAPG · S77
PLEKHM2 · S77
RPS8 · S77
RPL37A · S77
EIF1AD · S77
RPF2 · S77
RPL27A · S77
RPS10 · S77
GOLGA8K · S77
EXOSC8 · S77
−2.59
0
+2.59
Response (z)
−2.96
0
+2.96
Response (z)
−13
0
+13
Response (z)
ρ = 0.99
Var. ratio = 0.93
RPL9
RPL34
Predicted KO magnitude
Observed KO magnitude
ρ = 0.83
Var. ratio = 0.82
SNRPG
RRM1
Predicted KO magnitude
Observed KO magnitude
ρ = 0.71
Var. ratio = 0.017
0–5 zoom
phase
reporter
20 µm
SALL3 · S77
ELP2
TIPARP
Predicted KO magnitude
Observed KO magnitude
d
Observed
MLP
CatBoost
scPair
VAPA → OPS0021
MLP r = 0.80
TOMM70A → OPS0015
MLP r = 0.48
log(1 + response magnitude)
Observed KO rank
Observed KO rank
Ridge
GBDT
CatBoost
MLP
TabM
scButterfly
ResMLP
MultiTab
MIDAS
scPair
KO-response Pearson r
Magnitude Spearman ρ